Evaluating functions

1

a x **b** y

2 The value of the dependent variable when the independent variable is 3

3 $g\left(7\right)=-6$

4

a 2 **b** 0 **c** 1

5 46

6 -189

7

a 12 **b** -16

8

a 8 **b** -13

9

a $\frac{35}{4}$ **b** -7

10

a 1 **b** $-\frac{13}{5}$

11 $\frac{9}{2}$

12

a a^2+8a **b** b^2+8b

13

a 12 **b** m^2+8

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a

i $f(x) = 2 - \frac{1}{3}x$ ii 2

b

i $f(x) = \frac{6}{5}x + \frac{7}{5}$ i 5

c

i $f(x) = 3 - x - 6x^2$ ii -54

15

a $f(x) = \frac{1}{4}x - 2$ **b** 1

16

a

i $f(x) = \frac{7 - 5x}{-3}$ ii 1

b

i $f(x) = 4x^2 + x + 5$ ii 23

17 -10

18 -69

19

a True **b** False

20 $P(2) = 14\,000$, so after 2 years there are$14\,000$ fish in the lake.

Additional questions

21

a $W(19)$ represents the amount of money Sharon makes if she works 19 hours.

b The equation $W(n) = 165$ represents that Sharon makes $\$165$ after working n hours.

c $W(26) = \$390$

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22

a The equation $h(4) = 16$ represents that the tomato plant will be 16 inches tall after 4 weeks.

b 64 inches

23 Since $f(4) = 7$, we know that $4a + b = 7$. Since $f(-1) = -3$, we know that $-a + b = -3$. Solving this system of equations gives us $a = 2$ and $b = -1$.

24 An example scenario could be: Maggie has set aside 5 jelly beans for herself and will share 120 with her friends. Let x represent the number of people sharing the jelly beans (Maggie included) and let $f(x)$ represent the total number of jelly beans that Maggie ends up with.

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